ThetaSwap: Adverse Competition Oracle

The Risk Nobody Hedges

In a symmetric equilibrium, each of N LPs earns $1/N$ of fees [Ma and Crapis, 2024]. In practice, sophisticated actors concentrate fee revenue away from passive participants. This is **adverse competition** — a distinct, unhedged risk.

- **Impermanent loss (IL)** — price divergence between pool assets. Variance perpetuals perfectly hedge IL [Fukasawa et al., 2023]; Panoptic deploys this on-chain.
- **Loss-versus-rebalancing (LVR)** — adverse selection cost from informed traders. MEV-aware products address LVR.
- **Fee concentration risk** — the *competition structure* among LPs for fee revenue, captured by the FLAIR metric [Milionis et al., 2024]. **Nothing hedges this today.**

Research Question

Is there empirical demand for a hedging instrument against adverse competition? If so, how much are LPs willing to pay?

Data

ETH/USDC 30bps pool (Uniswap V3 mainnet)
41 days · 600 positions · 3,365 position-day observations · 597 exits

Treatment variable — concentration deviation:

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

- $A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$ fee concentration index (x_k = fee share, $\theta_k = 1/\text{blocklife}$)
- $A_T^{1/N} = \sqrt{\Theta/N^2}$ Ma–Crapis competitive null (equal shares $x_k = 1/N$)
- $\Delta^+ > 0$: fees more concentrated than the symmetric equilibrium

Model: The Shelter Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \boxed{\Delta_{t-\ell}} + \beta_2\, \Delta^2_{t-\ell} + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)$$

We expect $\beta_1 < 0$ (shelter effect):

- Mild fee concentration correlates with *high trading volume* — the pool is active, all LPs benefit from elevated fee revenue
- At low Δ, more concentration means more volume, which *lowers* exit probability
- This is **healthy competition**: sophisticated LPs attract flow that benefits everyone [Capponi et al., 2024]
- A linear-only model would conclude concentration is *protective* — missing the adverse regime entirely

Model: The Adverse Competition Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1\,\Delta_{t-\ell} + \beta_2\,\boxed{\Delta^2_{t-\ell}} + \beta_3\,\text{IL}_t + \beta_4\log(\text{age}_{i,t})\big)$$

We expect $\beta_2 > 0$ (adverse competition):

- Beyond a threshold, concentration no longer signals volume — it signals *fee extraction* by sophisticated actors
- Passive LP's effective fee rate $\hat{\phi}_i = x_i \cdot \phi_{\text{pool}}$ drops below exit threshold [Capponi and Zhu, 2024]
- The quadratic term Δ^2 captures this **nonlinear regime shift**: the marginal effect of concentration *flips sign* at the turning point
- Together: $\beta_1 < 0 + \beta_2 > 0 =$ **inverted-U** with turning point $\Delta^* = -\beta_1/(2\beta_2)$

Empirical Results

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	$\mathbf{-23.18}$	**0.012**	$\mathbf{+129.20}$	**0.030**	**0.090**
2	-43.42	*0.016*	$+226.92$	*0.065*	*0.096*
3	$\mathbf{-32.44}$	**0.001**	$\mathbf{+205.34}$	**0.004**	**0.079**

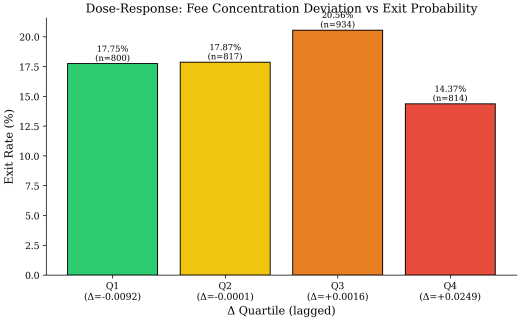
- **Lags 1 and 3**: both coefficients significant at 5%
- **Lag 2**: linear significant, quadratic marginal ($p = 0.065$)
- Signal **dissipates at lags 5–7** — LPs react to recent shocks

The Inverted-U

Two regimes:

1. **Shelter** ($\Delta < 0.09$)
   Mild concentration is protective — elevated volume benefits all LPs.

2. **Capponi** ($\Delta > 0.09$)
   Extreme concentration drives exits — passive LP fee rate drops below exit threshold.

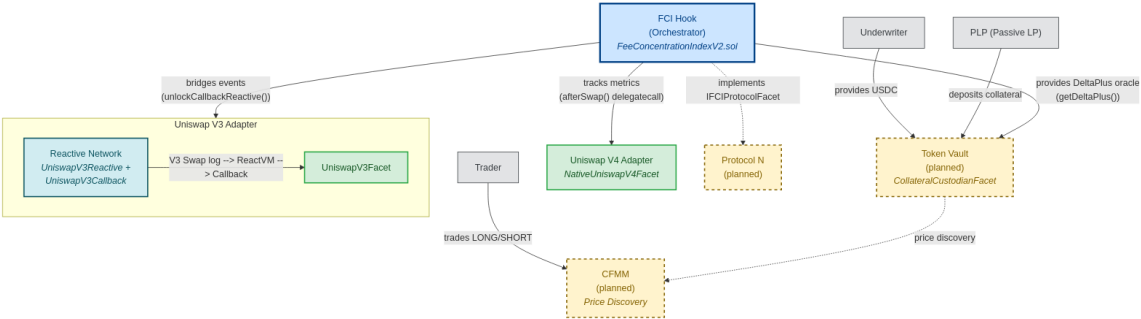


Turning point:

$$\Delta^* = \frac{-\beta_1}{2\beta_2} = \frac{23.18}{2 \times 129.20} = 0.090$$

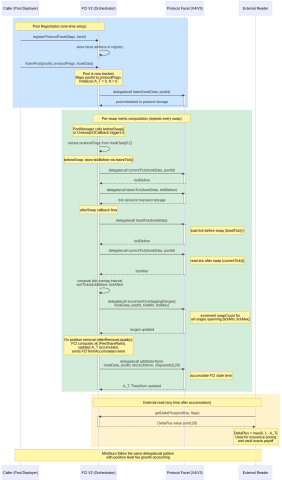
At $\Delta = 0.15$: marginal effect $= 2.27$

Architecture



FCI Hook at center — protocol adapters (V3, V4) feed swap/mint/burn events, Reactive Network bridges metrics cross-chain, Vault + CFMM settle downstream.

Cross-Protocol Modularity



`listenPool()` → swap/mint/burn events → metric update → Δ^+ derivation.

Live Demo

Run the integration test

```
forge test --match-path
"test/fee-concentration-index-v2/protocols/uniswapV4/
NativeV4FeeConcentrationIndex.integration.t.sol" -vv
```

What to observe:

1. Swap updates A_T — fee concentration index tracks each trade
2. Mint increments N — new positions enter the pool
3. Burn triggers Δ^+ derivation — exit computes concentration deviation
4. All metrics computed on-chain in a single V4 hook callback

What's Next

Built

- FCI oracle live on Uniswap V4
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value

Next

- Linearized power-squared CFMM (trading function complete in spec)
- Vault settlement mechanism

The mathematical specification is complete, the oracle tracks metrics across protocols, and the next milestone delivers the full insurance product.

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